Lesson 64: Mean-Field Games

AI/ML Mathematics

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1 Motivation and Overview

Consider a highway network carrying thousands of autonomous vehicles, each selecting routes and speeds to minimize its own travel time. Modeling this as an N-player stochastic game requires tracking the joint state of all vehicles – a state space that grows exponentially in N. Computing a Nash equilibrium in such a game is intractable for realistic population sizes. Yet the agents are *symmetric*: they share the same dynamics, the same action space, and the same reward structure, differing only in their individual states. Each driver does not care about the identity of the other vehicles, only about the *aggregate traffic density* on each road segment.

This symmetry suggests a radical simplification. Instead of tracking N individual states, we represent the collective behavior of all other agents by a single probability distribution – the *mean-field distribution* μ over states. In the limit $N \to \infty$, the law of large numbers ensures that the empirical distribution of agent states concentrates around a deterministic flow, and each individual agent faces what is effectively a single-agent MDP parameterized by this population distribution. The tools developed in this lesson – McKean–Vlasov dynamics, the forward-backward equilibrium system, and fixed-point existence theorems – will let us formalize this mean-field game (MFG) framework and connect it to practical multi-agent reinforcement learning algorithms.

Why Mean-Field Games Matter for RL/ML

- **Fleet management and ride-sharing.** Mean-field models enable scalable optimization of thousands of vehicles competing for passengers in ride-sharing platforms.
- **Crowd dynamics and evacuation.** MFGs model pedestrian flows where each individual optimizes against the aggregate crowd density.
- **Large-scale multi-agent RL.** Algorithms such as mean-field Q-learning [Yang et al., 2018] and mean-field actor-critic [Gu et al., 2021] scale MARL to populations of hundreds or thousands of agents.
- **Network congestion and resource allocation.** MFGs capture congestion effects in communication networks and shared computing resources where many identical agents compete for bandwidth.

Prerequisites. Stochastic game formalism and Markov perfect equilibrium (Lesson 61). Probability measures, weak convergence, and the simplex $\Delta(\mathcal{S})$ (Phase 01 measure theory, Lessons 13–14). Fixed-point theorems: Banach contraction mapping theorem and Schauder's fixed-point theorem (Lessons 15–16, Theorems on contractions and compact convex mappings). Bellman equations and value iteration for MDPs (Lesson 42).

Learning objectives. After completing this lesson, you will be able to:

- **Define** a mean-field game and **explain** how it arises as the infinite-population limit of a symmetric N-player stochastic game.
- **Derive** the McKean–Vlasov dynamics governing the evolution of the population distribution under a given policy.
- **Formulate** the forward-backward system that characterizes mean-field equilibria.
- **Prove** the existence of mean-field equilibria via Schauder's fixed-point theorem and **state** conditions for uniqueness via contraction arguments.
- **Describe** the mean-field Q-learning algorithm and **relate** it to the fixed-point iteration on the equilibrium operator.
- **Analyze** the quality of the mean-field approximation for finite N-player games, establishing the $O(1/\sqrt{N})$ approximation bound.

2 References

- Carmona and Delarue [2018a, Chapters 1–5] – Comprehensive treatment of mean-field game foundations, McKean–Vlasov equations, and existence theory in continuous time.
- Lasry and Lions [2007] – Original formulation of mean-field games via the coupled Hamilton–Jacobi–Bellman and Fokker–Planck PDE system.
- Huang et al. [2006] – Independent co-discovery of the mean-field game framework through the Nash certainty equivalence principle.
- Yang et al. [2018] – Mean-field multi-agent reinforcement learning: practical algorithms bridging MFG theory and deep RL.
- Cardaliaguet [2010] – Lecture notes providing an accessible introduction to mean-field games with complete proofs in the finite-state setting.

3 From N-Player Games to Mean-Field Limits

We begin by formalizing the class of N-player games for which the mean-field approximation is appropriate. The key structural requirement is *symmetry*: agents are interchangeable, and each agent's dynamics and rewards depend on the others only through the empirical distribution of their states.

Definition 3.1 (Symmetric N-player stochastic game)**.** A *symmetric N-player stochastic game* is a tuple $\mathcal{G}^N = (\mathcal{N}, \mathcal{S}, \mathcal{A}, P^N, R^N, \gamma)$ where:

1. $\mathcal{N} = \{1, \dots, N\}$ is the set of agents.
2. $\mathcal{S}$ is a finite state space shared by all agents. The joint state is $\mathbf{s} = (s_1, \dots, s_N) \in \mathcal{S}^N$.
3. $\mathcal{A}$ is a finite action space shared by all agents.
4. The transition kernel for agent i is
$$P^N(s_i' \mid s_i, a_i, \mathbf{s}_{-i}) = P^N\left(s_i' \,\middle|\, s_i, a_i, \frac{1}{N-1}\sum_{j\neq i}\delta_{s_j}\right),$$
depending on the other agents' states only through their empirical distribution.
5. The reward for agent i is
$$R^N(s_i, a_i, \mathbf{s}_{-i}) = R^N\left(s_i, a_i, \frac{1}{N-1}\sum_{j\neq i}\delta_{s_j}\right),$$
again depending on others only through the empirical measure.
6. $\gamma \in [0, 1)$ is the common discount factor.

The crucial feature is that the functions P^N and R^N depend on the other agents only through the *empirical measure* of their states. We denote this empirical measure explicitly.

Definition 3.2 (Empirical state distribution)**.** Given joint state $\mathbf{s} = (s_1, \dots, s_N) \in \mathcal{S}^N$, the *empirical state distribution* is
$$\mu^N(\cdot \mid \mathbf{s}) = \frac{1}{N}\sum_{i=1}^{N}\delta_{s_i}(\cdot) \in \Delta(\mathcal{S}),$$
where δ_{s_i} is the Dirac measure at s_i and $\Delta(\mathcal{S})$ is the probability simplex over $\mathcal{S}$. When excluding agent i, we write $\mu^{N,-i} = \frac{1}{N-1}\sum_{j\neq i}\delta_{s_j}$.

As the number of agents grows, the empirical measure concentrates around a deterministic limit.

Proposition 3.3 (Concentration of the empirical measure)**.** *Suppose the states* $(s_1, \ldots, s_N)$ *are drawn independently from a common distribution* $\mu \in \Delta(\mathcal{S})$*. Then for every* $s \in \mathcal{S}$ *and every* $\epsilon > 0$*,*

$$\mathbb{P}\big(\big|\mu^N(s) - \mu(s)\big| \geq \epsilon\big) \leq 2\exp\big(-2N\epsilon^2\big)\,.$$

Consequently, $\mu^N \to \mu$ *almost surely as* $N \to \infty$*.*

Proof. Fix $s \in \mathcal{S}$. Define $X_i = \mathbf{1}_{\{s_i = s\}}$ for $i = 1, \ldots, N$. Since the states are drawn independently from μ, the random variables $X_1, \ldots, X_N$ are i.i.d. with $\mathbb{E}[X_i] = \mu(s)$ and $X_i \in [0, 1]$. The empirical measure satisfies $\mu^N(s) = \frac{1}{N}\sum_{i=1}^N X_i$. By Hoeffding's inequality,

$$\mathbb{P}\big(\big|\mu^N(s) - \mu(s)\big| \geq \epsilon\big) \leq 2\exp\big(-2N\epsilon^2\big)\,.$$

Taking a union bound over the finite set $\mathcal{S}$ and applying the Borel–Cantelli lemma, $\|\mu^N - \mu\|_\infty \to 0$ almost surely as $N \to \infty$. □

Example 3.4 (Cars on a ring road)**.** Consider N cars on a circular road with $|\mathcal{S}| = K$ equally spaced positions. Each car chooses speed $a \in \mathcal{A} = \{0, 1, 2\}$ (stop, slow, fast). The transition probability depends on whether the next position is congested:

$$P^N(s' \mid s, a, \mu^{N,-i}) = \begin{cases} 1 - c \cdot \mu^{N,-i}(s') & \text{if } s' = s + a \pmod K, \\ c \cdot \mu^{N,-i}(s') & \text{if } s' = s \quad \text{(blocked)}, \end{cases}$$

where $c \in (0, 1)$ is a congestion parameter. In the limit $N \to \infty$, each car faces an MDP where the blocking probability at position s' is determined by the population density $\mu_t(s')$ rather than by the random empirical measure of finitely many other cars.

The example illustrates the fundamental insight: in the limit $N \to \infty$, each agent faces a single-agent optimization problem parameterized by the population distribution. We formalize this in the next section.

4 Mean-Field Game Formulation (Discrete Time)

Having motivated the transition from N-player games to population-level descriptions, we now define the mean-field game in discrete time.

Definition 4.1 (Discrete-time mean-field game)**.** A *discrete-time mean-field game* is a tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$ where:

1. $\mathcal{S}$ is a finite state space.
2. $\mathcal{A}$ is a finite action space.
3. $P : \mathcal{S} \times \mathcal{A} \times \Delta(\mathcal{S}) \to \Delta(\mathcal{S})$ is the transition kernel, where $P(s' \mid s, a, \mu)$ gives the probability of transitioning to s' when in state s, taking action a, and the population distribution is μ.
4. $R : \mathcal{S} \times \mathcal{A} \times \Delta(\mathcal{S}) \to \mathbb{R}$ is the reward function.
5. $\gamma \in [0, 1)$ is the discount factor.

The key distinction from a standard MDP is that both the transition kernel and the reward depend on the population distribution $\mu \in \Delta(\mathcal{S})$. For a fixed population flow, however, each agent faces a (time-varying) MDP.

Definition 4.2 (Mean-field MDP)**.** Given a population flow $(\mu_t)_{t\geq 0}$ with $\mu_t \in \Delta(\mathcal{S})$ for each t, the *mean-field MDP* is the single-agent MDP with time-varying transition kernel $P(\cdot \mid s, a, \mu_t)$ and time-varying reward $R(s, a, \mu_t)$. The value function of a policy π is

$$V_t^\pi(s; (\mu_\tau)_{\tau\geq t}) = \mathbb{E}\left[\sum_{k=0}^{\infty} \gamma^k R(s_{t+k}, a_{t+k}, \mu_{t+k}) \,\middle|\, s_t = s\right],$$

where $a_{t+k} \sim \pi(\cdot \mid s_{t+k})$ and $s_{t+k+1} \sim P(\cdot \mid s_{t+k}, a_{t+k}, \mu_{t+k})$.

When the population distribution is *stationary* ($\mu_t = \mu$ for all t), the mean-field MDP becomes a standard time-homogeneous MDP.

Proposition 4.3 (Bellman equation for the stationary mean-field MDP)**.** *For a fixed stationary population distribution $\mu \in \Delta(\mathcal{S})$, the optimal value function $V^*(s; \mu)$ satisfies the Bellman optimality equation*

$$V^*(s; \mu) = \max_{a\in\mathcal{A}} \left[R(s, a, \mu) + \gamma \sum_{s'\in\mathcal{S}} P(s' \mid s, a, \mu)\, V^*(s'; \mu)\right]. \tag{1}$$

The corresponding optimal policy is

$$\pi^*(s; \mu) \in \arg\max_{a\in\mathcal{A}} \left[R(s, a, \mu) + \gamma \sum_{s'\in\mathcal{S}} P(s' \mid s, a, \mu)\, V^*(s'; \mu)\right].$$

Proof. For fixed μ, the mean-field MDP is a standard discounted MDP with finite state and action spaces, transition kernel $P(\cdot \mid s, a, \mu)$, and reward $R(s, a, \mu)$. The Bellman optimality equation (1) follows directly from the standard MDP theory (Lesson 41, Theorem on Bellman optimality). Existence of an optimal deterministic stationary policy follows from the finiteness of $\mathcal{A}$. □

Mean-Field MDP as a Standard MDP

Since the mean-field MDP with fixed μ is a standard MDP, *all single-agent RL algorithms apply directly*: value iteration, policy iteration, Q-learning, and policy gradient methods can all be used to solve the agent's optimization problem for a given population distribution. The challenge of the mean-field game is that μ is itself determined by the agents' policies.

We now turn to the equation that governs the population distribution – the forward equation, or McKean–Vlasov dynamics.

5 McKean–Vlasov Dynamics

In the N-player game, the empirical state distribution evolves stochastically as agents take actions and transition between states. In the mean-field limit, this empirical distribution converges to a deterministic flow governed by the *McKean–Vlasov equation*.

Definition 5.1 (McKean–Vlasov dynamics (discrete time))**.** Given a policy $\pi : \mathcal{S} \to \Delta(\mathcal{A})$ and a population distribution $\mu_t \in \Delta(\mathcal{S})$, the *McKean–Vlasov equation* determines the next-step population distribution:

$$\mu_{t+1}(s') = \sum_{s\in\mathcal{S}} \sum_{a\in\mathcal{A}} P(s' \mid s, a, \mu_t)\, \pi(a \mid s)\, \mu_t(s), \qquad \forall\, s' \in \mathcal{S}. \tag{2}$$

Starting from an initial distribution μ_0, repeated application of (2) generates the *population flow* $(\mu_t)_{t\geq 0}$.

Remark 5.2. Equation (2) is the *forward equation* of the mean-field game. It is the discrete-time analogue of the Fokker–Planck (or Kolmogorov forward) equation in continuous time. The equation is nonlinear because the transition kernel P itself depends on μ_t.

We verify that the McKean–Vlasov dynamics preserve the simplex.

Lemma 5.3 (Simplex invariance)**.** *If $\mu_t \in \Delta(\mathcal{S})$ and $\pi(\cdot \mid s) \in \Delta(\mathcal{A})$ for all s, then μ_{t+1} defined by (2) satisfies $\mu_{t+1} \in \Delta(\mathcal{S})$.*

Proof. Non-negativity is immediate since P, π, and μ_t are all non-negative. For normalization:

$$\begin{aligned}\sum_{s' \in \mathcal{S}} \mu_{t+1}(s') &= \sum_{s'} \sum_{s} \sum_{a} P(s' \mid s, a, \mu_t)\, \pi(a \mid s)\, \mu_t(s) \\ &= \sum_{s} \sum_{a} \pi(a \mid s)\, \mu_t(s) \underbrace{\sum_{s'} P(s' \mid s, a, \mu_t)}_{=1} \\ &= \sum_{s} \mu_t(s) \underbrace{\sum_{a} \pi(a \mid s)}_{=1} = \sum_{s} \mu_t(s) = 1.\end{aligned}$$

□

We define the operator that maps a policy to its induced population flow, which will play a central role in the equilibrium analysis.

Definition 5.4 (Forward operator)**.** The *forward operator* Φ maps a policy π and an initial distribution μ_0 to the population flow:

$$\Phi(\pi, \mu_0) = (\mu_0, \mu_1, \mu_2, \ldots)$$

where μ_{t+1} is computed from μ_t and π via (2). In the stationary case, we seek a fixed distribution μ^* such that $\Phi(\pi, \mu^*) = (\mu^*, \mu^*, \ldots)$, i.e., μ^* is a *stationary distribution* of the McKean–Vlasov dynamics under π.

Definition 5.5 (Backward operator)**.** The *backward operator* Ψ maps a population flow $(\mu_t)_{t \geq 0}$ to the optimal policy for the corresponding mean-field MDP:

$$\Psi((\mu_t)_{t \geq 0}) = \pi^*,$$

where π^* is the optimal policy for the MDP with transitions $P(\cdot \mid s, a, \mu_t)$ and rewards $R(s, a, \mu_t)$. In the stationary case, $\Psi(\mu) = \pi^*(\cdot; \mu)$ solves the Bellman equation (1).

The forward and backward equations form a coupled system: the forward equation propagates the population distribution given a policy (Section 5), while the backward equation computes the optimal policy given a population flow (Section 4). A mean-field equilibrium requires solving both simultaneously.

Example 5.6 (Two-state McKean–Vlasov dynamics)**.** Let $\mathcal{S} = \{0, 1\}$, $\mathcal{A} = \{L, R\}$, and suppose the transition kernel is

$$P(1 \mid 0, R, \mu) = 1 - c \cdot \mu(1), \qquad P(0 \mid 1, L, \mu) = 1 - c \cdot \mu(0),$$

with $c = 0.5$, and all other transitions determined by normalization. Under a uniform random policy $\pi(L \mid s) = \pi(R \mid s) = 0.5$, starting from $\mu_0 = (0.5, 0.5)$, one step of the McKean–Vlasov dynamics gives

$$\begin{aligned}\mu_1(1) = \mu_0(0) \cdot \pi(R \mid 0) \cdot P(1 \mid 0, R, \mu_0) \\ + \mu_0(0) \cdot \pi(L \mid 0) \cdot P(1 \mid 0, L, \mu_0) \\ + \mu_0(1) \cdot \pi(R \mid 1) \cdot P(1 \mid 1, R, \mu_0) \\ + \mu_0(1) \cdot \pi(L \mid 1) \cdot P(1 \mid 1, L, \mu_0).\end{aligned}$$

With the specific transition model and $\mu_0(1) = 0.5$, this evaluates to a concrete number determined by the congestion parameter c. The distribution $\mu_1(0) = 1 - \mu_1(1)$ follows by normalization.

We now formalize the notion of equilibrium that couples the forward and backward equations.

6 Mean-Field Equilibrium

The central concept of mean-field game theory is the *mean-field equilibrium* (MFE), which requires the population flow and the individual optimal policy to be mutually consistent.

Definition 6.1 (Mean-field equilibrium – stationary case)**.** A pair (π^*, μ^*) is a *stationary mean-field equilibrium* (MFE) if:

1. **Optimality:** π^* is an optimal policy for the mean-field MDP with stationary population distribution μ^*, i.e., $\pi^* = \Psi(\mu^*)$.
2. **Consistency:** μ^* is the stationary distribution of the McKean–Vlasov dynamics under π^*, i.e.,
$$\mu^*(s') = \sum_{s\in\mathcal{S}} \sum_{a\in\mathcal{A}} P(s' \mid s, a, \mu^*)\, \pi^*(a \mid s)\, \mu^*(s), \qquad \forall\, s' \in \mathcal{S}.$$

Definition 6.2 (Mean-field equilibrium – general case)**.** A pair $(\pi^*, (\mu_t^*)_{t\geq 0})$ is a *mean-field equilibrium* if:

1. π^* is optimal for the mean-field MDP with population flow $(\mu_t^*)_{t\geq 0}$.
2. $(\mu_t^*)_{t\geq 0}$ is the McKean–Vlasov flow induced by π^* starting from initial distribution μ_0.

The equilibrium condition is a *fixed-point condition*. We make this precise by defining the combined equilibrium operator.

Definition 6.3 (Equilibrium operator)**.** The *equilibrium operator* $\Lambda : \Delta(\mathcal{S}) \to \Delta(\mathcal{S})$ (in the stationary case) is defined by the composition

$$\Lambda(\mu) = \Phi\big(\Psi(\mu),\, \mu\big),$$

where $\Psi(\mu) = \pi^*(\cdot; \mu)$ is the optimal policy for the mean-field MDP with distribution μ (backward step), and $\Phi(\pi^*, \mu)$ is the stationary distribution of the Markov chain with transition kernel

$$T(s' \mid s; \mu) = \sum_{a\in\mathcal{A}} P(s' \mid s, a, \mu)\, \pi^*(a \mid s; \mu)$$

(forward step). A stationary MFE is a fixed point of Λ: $\Lambda(\mu^*) = \mu^*$.

Mean-Field Equilibrium as a Fixed Point

A stationary mean-field equilibrium (π^*, μ^*) is characterized by the fixed-point equation

$$\mu^* = \Lambda(\mu^*),$$

where $\Lambda = \Phi \circ \Psi$ composes the backward operator (solve the MDP) with the forward operator (compute the stationary distribution). The existence and uniqueness of MFE reduce to fixed-point theory for the operator Λ on the compact convex set $\Delta(\mathcal{S})$.

Remark 6.4. The MFE is the infinite-population analogue of a symmetric Nash equilibrium: every agent plays the same policy π^*, and no agent has an incentive to deviate given the population distribution μ^*. The consistency condition replaces the finite-game requirement that each player's strategy be a best response to the others' strategies.

7 Existence and Uniqueness of Mean-Field Equilibria

We now establish the existence of MFE via topological fixed-point arguments and provide conditions for uniqueness via contraction mappings.

Assumption 7.1 (Regularity conditions)**.** Throughout this section we assume:

1. $\mathcal{S}$ and $\mathcal{A}$ are finite sets.
2. The transition kernel $P(s' \mid s, a, \mu)$ is continuous in $\mu \in \Delta(\mathcal{S})$ for every fixed (s, s', a).
3. The reward function $R(s, a, \mu)$ is continuous in μ for every fixed (s, a).
4. The Markov chain with transition kernel $T(s' \mid s; \mu) = \sum_a P(s' \mid s, a, \mu)\, \pi^*(a \mid s; \mu)$ is ergodic (irreducible and aperiodic) for every $\mu \in \Delta(\mathcal{S})$.

The ergodicity assumption ensures that the forward operator $\Phi(\pi, \cdot)$ has a unique stationary distribution for each policy, making Λ well-defined as a map on $\Delta(\mathcal{S})$.

Lemma 7.2 (Continuity of the backward operator)**.** *Under Assumption 7.1, the optimal value function $V^*(s; \mu)$ is continuous in μ for each s, and the optimal policy correspondence $\mu \mapsto \pi^*(\cdot; \mu)$ admits a continuous selection.*

Proof. For fixed μ, the Bellman operator $\mathcal{T}_\mu$ on $\mathbb{R}^{|\mathcal{S}|}$ is defined by

$$(\mathcal{T}_\mu V)(s) = \max_{a \in \mathcal{A}} \left[R(s, a, \mu) + \gamma \sum_{s'} P(s' \mid s, a, \mu)\, V(s') \right].$$

Since $\mathcal{T}_\mu$ is a γ-contraction on $(\mathbb{R}^{|\mathcal{S}|}, \|\cdot\|_\infty)$, its unique fixed point $V^*(\cdot; \mu)$ depends continuously on μ by the following argument. For any $\mu, \mu' \in \Delta(\mathcal{S})$,

$$\begin{aligned}
\|V^*(\cdot; \mu) - V^*(\cdot; \mu')\|_\infty &= \|\mathcal{T}_\mu V^*(\cdot; \mu) - \mathcal{T}_{\mu'} V^*(\cdot; \mu')\|_\infty \\
&\leq \|\mathcal{T}_\mu V^*(\cdot; \mu) - \mathcal{T}_\mu V^*(\cdot; \mu')\|_\infty \\
&\quad + \|\mathcal{T}_\mu V^*(\cdot; \mu') - \mathcal{T}_{\mu'} V^*(\cdot; \mu')\|_\infty \\
&\leq \gamma \|V^*(\cdot; \mu) - V^*(\cdot; \mu')\|_\infty + \epsilon(\mu, \mu'),
\end{aligned}$$

where $\epsilon(\mu, \mu') = \|\mathcal{T}_\mu V^*(\cdot; \mu') - \mathcal{T}_{\mu'} V^*(\cdot; \mu')\|_\infty \to 0$ as $\mu' \to \mu$ by continuity of R and P in μ. Rearranging, $\|V^*(\cdot; \mu) - V^*(\cdot; \mu')\|_\infty \leq \epsilon(\mu, \mu')/(1 - \gamma) \to 0$.

For the policy, when the optimal action is unique at every state (which holds generically), $\pi^*(a \mid s; \mu)$ is continuous in μ. In general, one can select a measurable (in fact continuous) optimal policy by Berge's maximum theorem, since the objective $Q^*(s, a; \mu) = R(s, a, \mu) + \gamma \sum_{s'} P(s' \mid s, a, \mu) V^*(s'; \mu)$ is continuous in μ and the constraint set $\mathcal{A}$ is finite (hence trivially continuous as a correspondence). □

Lemma 7.3 (Continuity of the forward operator)**.** *Under Assumption 7.1, the stationary distribution of the Markov chain with transition kernel $T(\cdot \mid \cdot; \mu)$ depends continuously on μ.*

Proof. By the ergodicity assumption, for each μ the chain has a unique stationary distribution $\nu(\mu) \in \Delta(\mathcal{S})$ satisfying $\nu(\mu)^\top T(\mu) = \nu(\mu)^\top$, where $T(\mu)$ is the $|\mathcal{S}| \times |\mathcal{S}|$ transition matrix. The entries of $T(\mu)$ are continuous in μ because P and π^* are continuous in μ (the latter by Lemma 7.2).

The stationary distribution is the left eigenvector of $T(\mu)$ corresponding to eigenvalue 1, normalized to sum to 1. Since $T(\mu)$ is ergodic, eigenvalue 1 is simple, and by standard perturbation theory for simple eigenvalues of matrices, the corresponding eigenvector depends continuously on the matrix entries. Hence $\nu(\mu)$ is continuous in μ. □

Theorem 7.4 (Existence of MFE via Schauder)**.** *Under Assumption 7.1, a stationary mean-field equilibrium exists.*

Proof. The equilibrium operator $\Lambda : \Delta(\mathcal{S}) \to \Delta(\mathcal{S})$ is defined by $\Lambda(\mu) = \Phi(\Psi(\mu), \mu)$, which maps μ to the stationary distribution of the Markov chain induced by the optimal policy $\Psi(\mu)$ and population distribution μ.

We verify the hypotheses of Schauder's fixed-point theorem (Lesson 16, Theorem on fixed points for continuous maps on compact convex sets):

1. **Domain:** $\Delta(\mathcal{S}) \subset \mathbb{R}^{|\mathcal{S}|}$ is compact and convex.
2. **Self-mapping:** By Lemma 5.3 and the uniqueness of stationary distributions under ergodicity, $\Lambda(\mu) \in \Delta(\mathcal{S})$ for every $\mu \in \Delta(\mathcal{S})$.
3. **Continuity:** $\Lambda = \Phi \circ \Psi$ is the composition of two continuous maps. The backward operator Ψ is continuous by Lemma 7.2, and the forward operator Φ (mapping policy to stationary distribution) is continuous by Lemma 7.3.

Since Λ is a continuous self-map of the compact convex set $\Delta(\mathcal{S}) \subset \mathbb{R}^{|\mathcal{S}|}$, Schauder's fixed-point theorem guarantees the existence of $\mu^* \in \Delta(\mathcal{S})$ with $\Lambda(\mu^*) = \mu^*$. Setting $\pi^* = \Psi(\mu^*)$, the pair (π^*, μ^*) is a stationary MFE by Definition 6.1. □

Existence of Mean-Field Equilibrium

Under continuity and ergodicity assumptions (Assumption 7.1), every discrete-time mean-field game with finite state and action spaces admits at least one stationary mean-field equilibrium. The proof relies on Schauder's fixed-point theorem applied to the equilibrium operator $\Lambda : \Delta(\mathcal{S}) \to \Delta(\mathcal{S})$.

We now turn to uniqueness, which requires stronger conditions.

Assumption 7.5 (Lipschitz conditions)**.** There exist constants $L_P, L_R > 0$ such that for all $s, s' \in \mathcal{S}$, $a \in \mathcal{A}$, and $\mu, \mu' \in \Delta(\mathcal{S})$:

$$\begin{aligned} |P(s' \mid s, a, \mu) - P(s' \mid s, a, \mu')| &\le L_P \|\mu - \mu'\|_1, \\ |R(s, a, \mu) - R(s, a, \mu')| &\le L_R \|\mu - \mu'\|_1. \end{aligned}$$

Theorem 7.6 (Uniqueness via contraction)**.** *Under Assumptions 7.1–7.5, suppose the discount factor satisfies*

$$\gamma \cdot \left(L_P \cdot \frac{R_{\max}}{(1-\gamma)^2} + \frac{L_P + L_R}{1-\gamma} \right) \cdot C_T < 1, \tag{3}$$

where $R_{\max} = \max_{s,a,\mu} |R(s, a, \mu)|$ and C_T is the sensitivity constant of the stationary distribution with respect to transition matrix perturbations. Then Λ is a contraction on $(\Delta(\mathcal{S}), \|\cdot\|_1)$, and the MFE is unique.

Proof. We bound the Lipschitz constant of Λ by bounding the sensitivity of each component.

Step 1: Sensitivity of the value function. From the proof of Lemma 7.2, for any μ, μ':

$$\|V^*(\cdot; \mu) - V^*(\cdot; \mu')\|_\infty \le \frac{1}{1-\gamma} \left(L_R + \frac{\gamma\, L_P\, R_{\max}}{1-\gamma} \right) \|\mu - \mu'\|_1.$$

This follows by bounding $\epsilon(\mu, \mu')$ in terms of the Lipschitz constants. Specifically,

$$\begin{aligned} |(\mathcal{T}_\mu V)(s) - (\mathcal{T}_{\mu'} V)(s)| &\le \max_a \big[|R(s, a, \mu) - R(s, a, \mu')| \\ &\quad + \gamma |P(\cdot|s, a, \mu) - P(\cdot|s, a, \mu')| \cdot \|V\|_\infty \big] \\ &\le L_R \|\mu - \mu'\|_1 + \gamma\, L_P \|V\|_\infty \|\mu - \mu'\|_1. \end{aligned}$$

Setting $V = V^*(\cdot;\mu')$ with $\|V^*(\cdot;\mu')\|_\infty \le R_{\max}/(1-\gamma)$ gives the bound.

Step 2: Sensitivity of the optimal policy. The Q-function $Q^*(s,a;\mu)$ inherits the Lipschitz property from V^*. When the optimal action is unique, the policy is locally constant and hence Lipschitz with constant 0. In general, by considering soft-max policies (or policies that break ties consistently), the policy varies Lipschitz-continuously in μ with a constant controlled by the value function sensitivity.

Step 3: Sensitivity of the stationary distribution. By standard perturbation theory for ergodic Markov chains [Bensoussan et al., 2013, Chapter 3], the stationary distribution ν of a transition matrix T satisfies

$$\|\nu(T) - \nu(T')\|_1 \le C_T \|T - T'\|_\infty,$$

where C_T depends on the spectral gap of the chain. The perturbation $\|T(\mu) - T(\mu')\|_\infty$ is bounded by the combined sensitivity of P and π^* to μ.

Step 4: Composing the bounds. Combining Steps 1–3:

$$\|\Lambda(\mu) - \Lambda(\mu')\|_1 \le \underbrace{\gamma \cdot \left(L_P \cdot \frac{R_{\max}}{(1-\gamma)^2} + \frac{L_P + L_R}{1-\gamma} \right) \cdot C_T}_{=:\,\kappa} \|\mu - \mu'\|_1.$$

When $\kappa < 1$ (condition (3)), Λ is a contraction. By the Banach fixed-point theorem, the fixed point is unique. □

Non-Uniqueness of Mean-Field Equilibria

Without the contraction condition (3), multiple MFEs can exist. This is analogous to the multiplicity of Nash equilibria in finite games. In particular, when the discount factor is large (agents are patient) or the mean-field dependence is strong (L_P, L_R are large), the equilibrium operator Λ may have multiple fixed points. Example **??** below illustrates this phenomenon.

Example 7.7 (Congestion game with explicit MFE)**.** Consider $\mathcal{S} = \{0,1\}$ (two routes), $\mathcal{A} = \{0,1\}$ (stay or switch), and the congestion-dependent reward $R(s,a,\mu) = 1-\mu(s)$ (higher reward when the current route is less congested). Transitions: $P(1-s \mid s,1,\mu) = 1$ (switching always succeeds) and $P(s \mid s,0,\mu) = 1$ (staying is deterministic).

With discount γ close to 0, agents are myopic and the MFE approaches the static Nash equilibrium. If agents are symmetric with initial $\mu_0 = (0.5, 0.5)$, the unique stationary MFE is $\mu^* = (0.5, 0.5)$ with the optimal policy "stay" at each state (since switching does not improve the reward when both routes are equally congested). We verify:

- **Optimality:** With $\mu^* = (0.5, 0.5)$, the reward for staying is $1 - 0.5 = 0.5$ and the reward for switching is also $1 - 0.5 = 0.5$ (since both routes have density 0.5). Any policy is optimal, including "stay."
- **Consistency:** Under "stay," $\mu_{t+1} = \mu_t = (0.5, 0.5)$, confirming stationarity.

8 Mean-Field Q-Learning

The fixed-point characterization of MFE suggests a natural computational approach: alternate between solving the backward equation (optimizing the agent's policy given the population) and the forward equation (updating the population given the policy). This is the basis of *mean-field Q-learning* [Yang et al., 2018].

Definition 8.1 (Mean-field Q-function)**.** The *mean-field Q-function* is defined as

$$Q(s,a,\mu) = R(s,a,\mu) + \gamma \sum_{s' \in \mathcal{S}} P(s' \mid s,a,\mu)\, V(s',\mu'), \tag{4}$$

where $V(s,\mu) = \max_{a\in\mathcal{A}} Q(s,a,\mu)$ and μ' is the next-step population distribution consistent with the current policy derived from Q.

Definition 8.2 (Mean-field Q-learning algorithm)**.** The *mean-field Q-learning* algorithm alternates two steps at each iteration $k = 0, 1, 2, \ldots$:

Step 1 (Q-update): With the current population distribution μ_k held fixed, perform a Q-learning update:

$$Q_{k+1}(s,a,\mu_k) = (1-\alpha_k)\, Q_k(s,a,\mu_k) + \alpha_k \Big[R(s,a,\mu_k) + \gamma \max_{a'} Q_k(s',a',\mu_k) \Big], \tag{5}$$

where $\alpha_k \in (0,1)$ is the learning rate and $s' \sim P(\cdot \mid s,a,\mu_k)$.

Step 2 (Mean-field update): Update the population distribution using a Boltzmann policy derived from the current Q-function:

$$\mu_{k+1}(s') = \sum_{s\in\mathcal{S}} \sum_{a\in\mathcal{A}} P(s' \mid s,a,\mu_k)\, \sigma_\beta(a \mid s,\mu_k)\, \mu_k(s), \tag{6}$$

where $\sigma_\beta(a \mid s,\mu) = \frac{\exp(\beta\, Q_k(s,a,\mu))}{\sum_{a'} \exp(\beta\, Q_k(s,a',\mu))}$ is the Boltzmann (soft-max) policy with inverse temperature $\beta > 0$.

Proposition 8.3 (Fixed-point interpretation)**.** *The mean-field Q-learning algorithm is a form of fixed-point iteration on the equilibrium operator Λ. Specifically:*

- *Step 1 approximates the backward operator Ψ: given μ_k, it computes the optimal Q-function (and hence the optimal policy) for the mean-field MDP with distribution μ_k.*
- *Step 2 approximates the forward operator Φ: given the policy σ_β derived from Q_k, it advances the population distribution by one step of the McKean–Vlasov dynamics.*

In the limit $\beta \to \infty$, the Boltzmann policy converges to the greedy policy, and the algorithm reduces to exact fixed-point iteration on Λ.

Proof. At convergence ($Q_k \to Q^*$, $\mu_k \to \mu^*$), Step 1 yields the fixed point of the Bellman operator for the mean-field MDP with distribution μ^*, so Q^* encodes the optimal policy $\pi^* = \Psi(\mu^*)$. Step 2 with $\beta \to \infty$ reduces to the McKean–Vlasov equation under π^*, so μ^* is the stationary distribution of the forward dynamics under π^*, i.e., $\mu^* = \Phi(\pi^*, \mu^*)$. Together, $\mu^* = \Lambda(\mu^*)$, which is the MFE fixed-point condition. □

Proposition 8.4 (Convergence of mean-field Q-learning)**.** *Under Assumptions 7.1–7.5 and the contraction condition (3), suppose the learning rates satisfy the Robbins–Monro conditions $\sum_k \alpha_k = \infty$, $\sum_k \alpha_k^2 < \infty$, and the inverse temperature $\beta_k \to \infty$ sufficiently slowly. Then the iterates (Q_k, μ_k) of the mean-field Q-learning algorithm converge to the unique MFE (Q^*, μ^*) almost surely.*

Proof sketch. The convergence analysis proceeds in two stages following the two-timescale stochastic approximation framework (Lesson 19, SA theory).

Fast timescale (Q-update): For fixed μ_k, Step 1 is standard tabular Q-learning on a finite MDP. Under the Robbins–Monro conditions on α_k and appropriate exploration, $Q_k(\cdot,\cdot,\mu_k) \to Q^*(\cdot,\cdot,\mu_k)$ by the convergence theorem for Q-learning (Lesson 50).

Slow timescale (mean-field update): Viewing Q_k as approximately converged (the fast timescale has equilibrated), Step 2 becomes deterministic iteration of Λ (with a vanishing Boltzmann smoothing error as $\beta_k \to \infty$). Under the contraction condition (3), Λ is a contraction, so $\mu_k \to \mu^*$.

The two-timescale analysis of Bensoussan et al. [2013, Chapter 6] formalizes the separation of timescales and establishes the almost sure convergence of the joint iterates. □

From Theory to Practice in Mean-Field MARL

- **Scalability.** Mean-field Q-learning scales to hundreds of agents because each agent maintains a Q-function of size $|\mathcal{S}| \times |\mathcal{A}|$ rather than $|\mathcal{S}|^N \times |\mathcal{A}|^N$.
- **Function approximation.** In practice, $Q(s, a, \mu)$ is parameterized by a neural network that takes state, action, and a representation of μ as inputs, enabling generalization across population distributions.
- **Decentralized execution.** Once the MFE policy π^* is learned, each agent can execute independently using only its local state, since the policy does not depend on other agents' identities.

Example 8.5 (Mean-field Q-learning on the congestion game)**.** Returning to Example 7.7 with $\mathcal{S} = \{0, 1\}$, $\mathcal{A} = \{0, 1\}$, $R(s, a, \mu) = 1 - \mu(s)$, and deterministic transitions. Initialize $Q_0(s, a, \mu_0) = 0$ and $\mu_0 = (0.6, 0.4)$. At iteration $k = 0$:

Step 1: The immediate reward under $\mu_0 = (0.6, 0.4)$ is $R(0, \cdot, \mu_0) = 1 - 0.6 = 0.4$ and $R(1, \cdot, \mu_0) = 1 - 0.4 = 0.6$. With $\alpha_0 = 1$ and $Q_0 = 0$, one full update gives $Q_1(s, a, \mu_0) = R(s, a, \mu_0)$.

Step 2: With $\beta = 5$, the Boltzmann policy at state 0 favors switching (since $R(0, 1, \mu_0) = 0.6 > 0.4 = R(0, 0, \mu_0)$, and after switching the agent moves to state 1). The population shifts toward μ_1 with more mass on state 1.

Subsequent iterations continue to equalize the population across states, converging to $\mu^* = (0.5, 0.5)$.

9 Connection to N-Player Games

The mean-field game is useful only insofar as its equilibria approximate the behavior of large but finite populations. The following theorem makes this connection precise: an MFE yields an approximate Nash equilibrium for the N-player game with approximation error $O(1/\sqrt{N})$.

Definition 9.1 (ϵ-Nash equilibrium)**.** A symmetric policy profile $(\pi, \ldots, \pi)$ is an *ϵ-Nash equilibrium* of the N-player game $\mathcal{G}^N$ if for every agent i and every alternative policy $\tilde{\pi}$:

$$V_i^{(\tilde{\pi}, \pi_{-i})}(s_i) \leq V_i^{(\pi, \ldots, \pi)}(s_i) + \epsilon, \qquad \forall\, s_i \in \mathcal{S},$$

where $V_i^{(\cdot)}$ is agent i's discounted value under the given policy profile.

Theorem 9.2 (N-player approximation by MFE)**.** *Let (π^*, μ^*) be a stationary MFE for the mean-field game $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$, and let $\mathcal{G}^N$ be the corresponding N-player symmetric stochastic game (Definition 3.1). Under Assumption 7.5, if agents' initial states are drawn i.i.d. from μ^*, then the symmetric profile $(\pi^*, \ldots, \pi^*)$ is an ϵ_N-Nash equilibrium of $\mathcal{G}^N$ with*

$$\epsilon_N = \frac{C}{\sqrt{N}},$$

where $C = \frac{2(L_R + \gamma\, L_P\, R_{\max}/(1-\gamma))}{(1-\gamma)} \cdot |\mathcal{S}|^{1/2}$ depends on the Lipschitz constants and the problem parameters but not on N.

Proof. Fix agent i and suppose all other agents $j \neq i$ use π^*. Since the initial states are drawn i.i.d. from μ^* and all agents use π^*, the empirical measure $\mu_t^{N,-i} = \frac{1}{N-1} \sum_{j \neq i} \delta_{s_j(t)}$ of other agents' states at time t deviates from the mean-field flow μ^* (which is stationary under π^*).

Step 1: Concentration bound. At time t, conditional on the history, the states $s_j(t)$ for $j \neq i$ are not independent (they depend on the shared empirical measure at previous times).

However, a propagation-of-chaos argument [Carmona and Delarue, 2018a, Chapter 4] shows that the dependence weakens as $N \to \infty$, and the empirical measure concentrates:

$$\mathbb{E}\Big[\|\mu_t^{N,-i} - \mu^*\|_1\Big] \le \frac{C_1|\mathcal{S}|^{1/2}}{\sqrt{N}}, \tag{7}$$

where C_1 depends on L_P and T but not on N. For the stationary case, the bound holds uniformly over all t.

Step 2: Value perturbation. Agent i's value in the N-player game under $(\pi^*, \dots, \pi^*)$ differs from the mean-field value $V^{\pi^*}(s_i; \mu^*)$ by

$$\begin{aligned}
&\Big|V_i^{(\pi^*,\dots,\pi^*)}(s_i) - V^{\pi^*}(s_i;\mu^*)\Big| \\
&\le \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t \Big|R(s_t, a_t, \mu_t^{N,-i}) - R(s_t, a_t, \mu^*)\Big|\right] \\
&\quad + \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t (\text{transition perturbation terms})\right] \\
&\le \frac{1}{1-\gamma}\left(L_R + \frac{\gamma\, L_P\, R_{\max}}{1-\gamma}\right) \cdot \frac{C_1|\mathcal{S}|^{1/2}}{\sqrt{N}}.
\end{aligned}$$

The reward perturbation uses Assumption 7.5 and the concentration bound (7). The transition perturbation propagates similarly through the Bellman recursion.

Step 3: Deviation bound. Now suppose agent i deviates to an arbitrary policy $\tilde{\pi}$. Since agent i is a single agent among N, its deviation changes the empirical measure $\mu^{N,-i}$ by at most $O(1/N)$, which is dominated by the $O(1/\sqrt{N})$ fluctuation. Agent i's value under $\tilde{\pi}$ in the N-player game is close to its value under $\tilde{\pi}$ in the mean-field MDP with distribution μ^*. But π^* is optimal for the mean-field MDP, so

$$V^{\tilde{\pi}}(s_i;\mu^*) \le V^{\pi^*}(s_i;\mu^*).$$

Combining with Step 2 applied to both π^* and $\tilde{\pi}$:

$$V_i^{(\tilde{\pi},\pi^*_{-i})}(s_i) \le V_i^{(\pi^*,\dots,\pi^*)}(s_i) + \frac{2C_1(L_R + \gamma L_P R_{\max}/(1-\gamma))}{(1-\gamma)\sqrt{N}}.$$

Setting $C = 2C_1(L_R + \gamma L_P R_{\max}/(1-\gamma))/(1-\gamma) \cdot |\mathcal{S}|^{1/2}$ gives $\epsilon_N = C/\sqrt{N}$. □

Quality of the Mean-Field Approximation

A mean-field equilibrium (π^*, μ^*) provides an $O(1/\sqrt{N})$-Nash equilibrium for the corresponding N-player symmetric stochastic game. This justifies the mean-field approach for large populations: the approximation improves as the number of agents increases, and the computational cost is independent of N.

Remark 9.3. The $O(1/\sqrt{N})$ rate is generally tight. It matches the central limit theorem rate for fluctuations of the empirical measure around its mean. For certain structured problems (e.g., linear-quadratic MFGs), the rate can be improved to $O(1/N)$ using more refined concentration arguments [Carmona and Delarue, 2018a, Chapter 5].

10 Exercises

Exercise 10.1 (McKean–Vlasov dynamics for a two-state game)**.** $(*)$
Consider the mean-field game with $\mathcal{S} = \{0, 1\}$, $\mathcal{A} = \{L, R\}$, transition kernel

$$P(1 \mid 0, R, \mu) = 0.8 - 0.3\,\mu(1), \qquad P(0 \mid 1, L, \mu) = 0.7 - 0.2\,\mu(0),$$

and $P(s \mid s, a, \mu) = 1 - P(1 - s \mid s, a, \mu)$ for all other cases. Starting from $\mu_0 = (0.4, 0.6)$ and the policy $\pi(R \mid 0) = 0.7$, $\pi(L \mid 1) = 0.6$, compute μ_1 and μ_2 using the McKean–Vlasov equation (2).

Exercise 10.2 (Verifying a mean-field equilibrium)**.** $(*)$
Consider the congestion game from Example 7.7 with $R(s, a, \mu) = 1 - \mu(s)$, deterministic transitions (stay or switch), and $\gamma = 0.9$. Verify that $(\pi^*, \mu^*) = (\text{“stay”}, (0.5, 0.5))$ is an MFE by checking both the optimality and consistency conditions of Definition 6.1. Is this the unique MFE?

Exercise 10.3 (Schauder existence for finite-horizon MFG)**.** $(**)$
Consider a finite-horizon mean-field game over T periods with finite $\mathcal{S}$ and $\mathcal{A}$. Define the equilibrium operator $\Lambda_T : \Delta(\mathcal{S})^T \to \Delta(\mathcal{S})^T$ that maps a population flow $(\mu_0, \ldots, \mu_{T-1})$ to the flow induced by backward induction (compute optimal policies via dynamic programming) followed by forward propagation (McKean–Vlasov dynamics). Under continuity assumptions on P and R, prove that Λ_T has a fixed point.

Hint: $\Delta(\mathcal{S})^T \subset \mathbb{R}^{|\mathcal{S}| \cdot T}$ is compact and convex. Verify that Λ_T is continuous (use the continuity of finite-horizon dynamic programming in the reward and transition parameters) and apply Brouwer's fixed-point theorem.

Exercise 10.4 (Deriving the mean-field Q-learning update)**.** $(**)$
Starting from the mean-field Bellman equation

$$Q^*(s, a, \mu) = R(s, a, \mu) + \gamma \sum_{s'} P(s' \mid s, a, \mu) \max_{a'} Q^*(s', a', \mu'),$$

derive the stochastic approximation update (5) by replacing the expected next-state value with a sample. State the Robbins–Monro conditions on (α_k) that guarantee convergence for fixed μ, and explain why the Boltzmann mean-field update (6) is a natural companion step.

Exercise 10.5 (Approximation bound for finite populations)**.** $(***)$
Let (π^*, μ^*) be a stationary MFE. Suppose N agents each use π^* with initial states drawn i.i.d. from μ^*. Prove that for any single agent i and any deviation policy $\tilde{\pi}$:

$$V_i^{(\tilde{\pi}, \pi^*_{-i})}(s_i) - V_i^{(\pi^*, \ldots, \pi^*)}(s_i) \leq \frac{C}{\sqrt{N}}$$

for a constant C independent of N. You may assume the Lipschitz conditions of Assumption 7.5 and the concentration bound (7).

Hint: Bound the value difference by decomposing into (i) the gap between the N-player value and the mean-field value for both π^* and $\tilde{\pi}$, and (ii) the optimality of π^* in the mean-field MDP.

Exercise 10.6 (Constructing an MFG with multiple equilibria)**.** $(***)$
Construct a mean-field game with $\mathcal{S} = \{0, 1\}$, $\mathcal{A} = \{0, 1\}$, and $\gamma = 0.95$ that admits at least two distinct stationary MFEs. Specify $P(s' \mid s, a, \mu)$ and $R(s, a, \mu)$ explicitly. Verify each MFE by checking both optimality and consistency.

Hint: Use a reward function with a coordination component, e.g., $R(s, a, \mu) = s \cdot \mu(1) + (1 - s) \cdot \mu(0)$ (agents prefer to be in the majority), to create a bistable system.

11 Summary

- A *symmetric N-player stochastic game* is one where agents are exchangeable and each agent's transitions and rewards depend on the others only through the empirical state distribution $\mu^N = \frac{1}{N} \sum_i \delta_{s_i}$.

- As $N \to \infty$, the empirical measure concentrates around a deterministic population distribution μ, reducing the N-player game to a *mean-field game* where each agent solves a single-agent MDP parameterized by μ.
- The *McKean–Vlasov equation* is the forward equation governing the evolution of μ_t under a given policy: $\mu_{t+1}(s') = \sum_{s,a} P(s'|s, a, \mu_t)\, \pi(a|s)\, \mu_t(s)$.
- A *mean-field equilibrium* (MFE) is a pair (π^*, μ^*) where π^* is optimal for the mean-field MDP with distribution μ^*, and μ^* is the stationary distribution of the McKean–Vlasov dynamics under π^*.
- The MFE is characterized as a fixed point of the equilibrium operator $\Lambda = \Phi \circ \Psi$, composing the backward operator (solve MDP) with the forward operator (compute stationary distribution).
- *Existence* of MFE follows from Schauder's fixed-point theorem: Λ is a continuous self-map of the compact convex set $\Delta(\mathcal{S})$.
- *Uniqueness* holds when the discount factor is small enough and the mean-field coupling is weak enough to make Λ a contraction.
- *Mean-field Q-learning* alternates between Q-learning (backward step) and McKean–Vlasov updates (forward step), implementing fixed-point iteration on Λ with convergence guaranteed under contraction conditions.
- An MFE provides an $O(1/\sqrt{N})$-Nash equilibrium for the N-player game, justifying the mean-field approximation for large populations.
- The computational cost of solving the mean-field game is *independent of* N, making it the only tractable approach for very large populations.

12 Further Reading

- **Continuous-time formulation.** In continuous time, the McKean–Vlasov equation becomes the Fokker–Planck PDE and the backward equation becomes the Hamilton–Jacobi–Bellman PDE. The coupled HJB-FP system is the classical formulation of Lasry and Lions [2007] and Huang et al. [2006]. See Cardaliaguet [2010] for lecture notes with complete proofs.
- **Common noise and the master equation.** Carmona and Delarue [2018b] extends the theory to settings where agents share a common source of randomness (e.g., weather affecting all vehicles). The *master equation* is a PDE on the space of probability measures that encodes the entire MFG solution.
- **Connections to optimal transport.** The McKean–Vlasov dynamics can be viewed through the lens of optimal transport theory, where the population flow minimizes a cost functional over paths in the space of measures. See Carmona and Delarue [2018a, Chapter 5].
- **Mean-field MARL algorithms.** Beyond mean-field Q-learning [Yang et al., 2018], recent work includes mean-field actor-critic methods [Gu et al., 2021] and decentralized approaches. Zhang et al. [2021] provides a comprehensive survey of multi-agent RL algorithms, including mean-field methods.

- **Forward to Lesson 65 (Communication in MARL).** Lesson 65 studies communication protocols in multi-agent systems, complementing the mean-field approach with explicit information exchange between agents. While MFGs assume agents interact only through the population distribution, communication-based methods allow richer agent interactions.
- **Historical notes.** Mean-field games were independently introduced by Lasry and Lions [2007] (Lasry and Lions, mathematicians) and Huang et al. [2006] (Huang, Malhamé, and Caines, engineers) in 2006–2007. The term "mean field" is borrowed from statistical physics, where it describes the approximation of particle interactions by an average field.

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